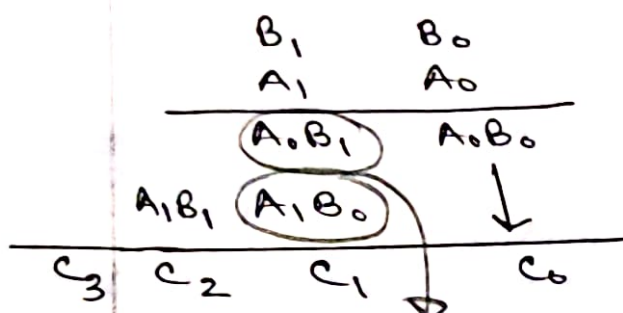


## Binary Multiplier

$$\begin{array}{r}
 23 \\
 \times 21 \\
 \hline
 23 \\
 460 \\
 \hline
 483
 \end{array}$$

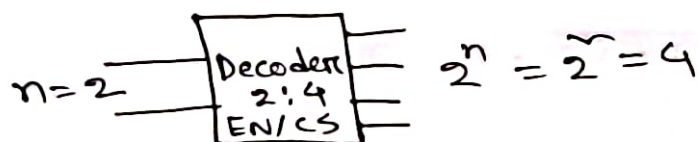


2-bit Add ~~2010~~, half Adder ~~2010~~

\* by changing the bit pattern.

29.11.2022

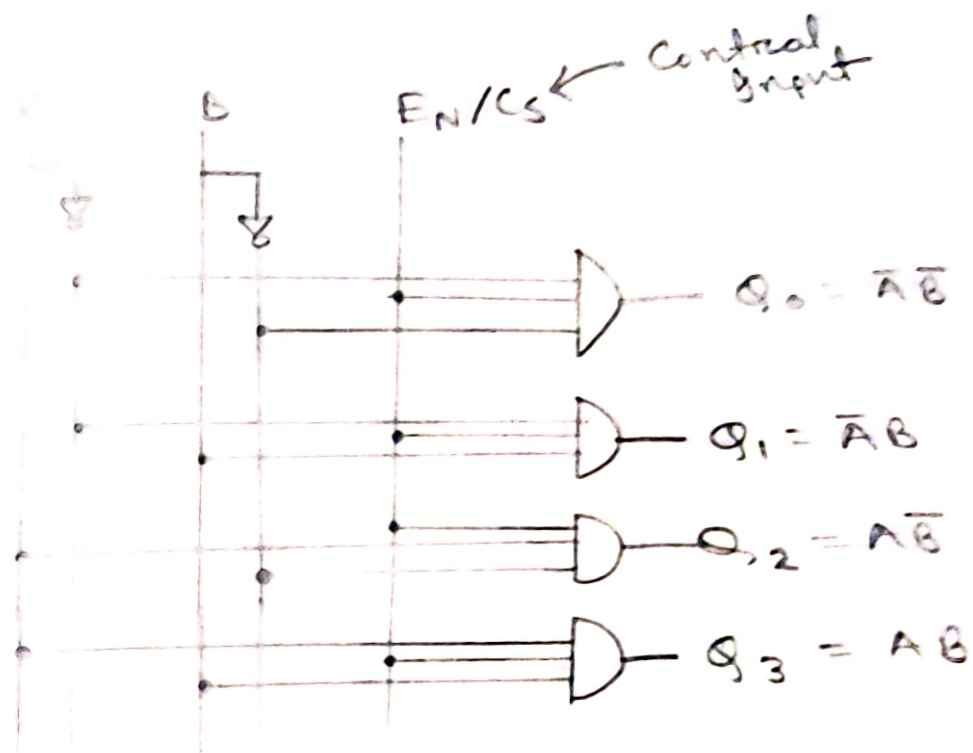
## Decoder



Bubble ~~2010~~ 0, otherwise 1

En = Enable

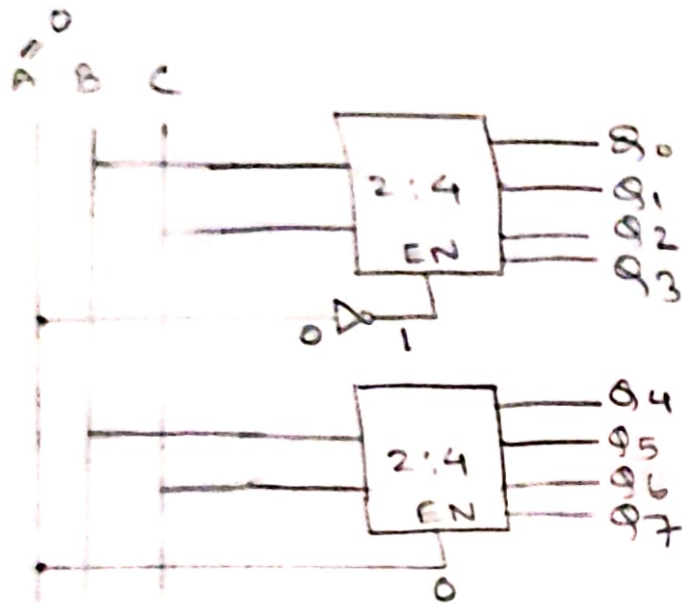
Cs = chip select



| Input |   |   | Output |       |       |       |
|-------|---|---|--------|-------|-------|-------|
| $C_s$ | A | B | $Q_3$  | $Q_2$ | $Q_1$ | $Q_0$ |
| 1     | 0 | 0 | 0      | 0     | 0     | 1     |
| 1     | 0 | 1 | 0      | 0     | 1     | 0     |
| 1     | 1 | 0 | 0      | 1     | 0     | 0     |
| 1     | 1 | 1 | 1      | 0     | 0     | 0     |
| 0     | X | X | 0      | 0     | 0     | 0     |

control input, প্রত্যেকটি AND gate-র  
 আর short কালের সহায় থাকবে।

# Design 3:8 Decoder using 2:4 decoder.

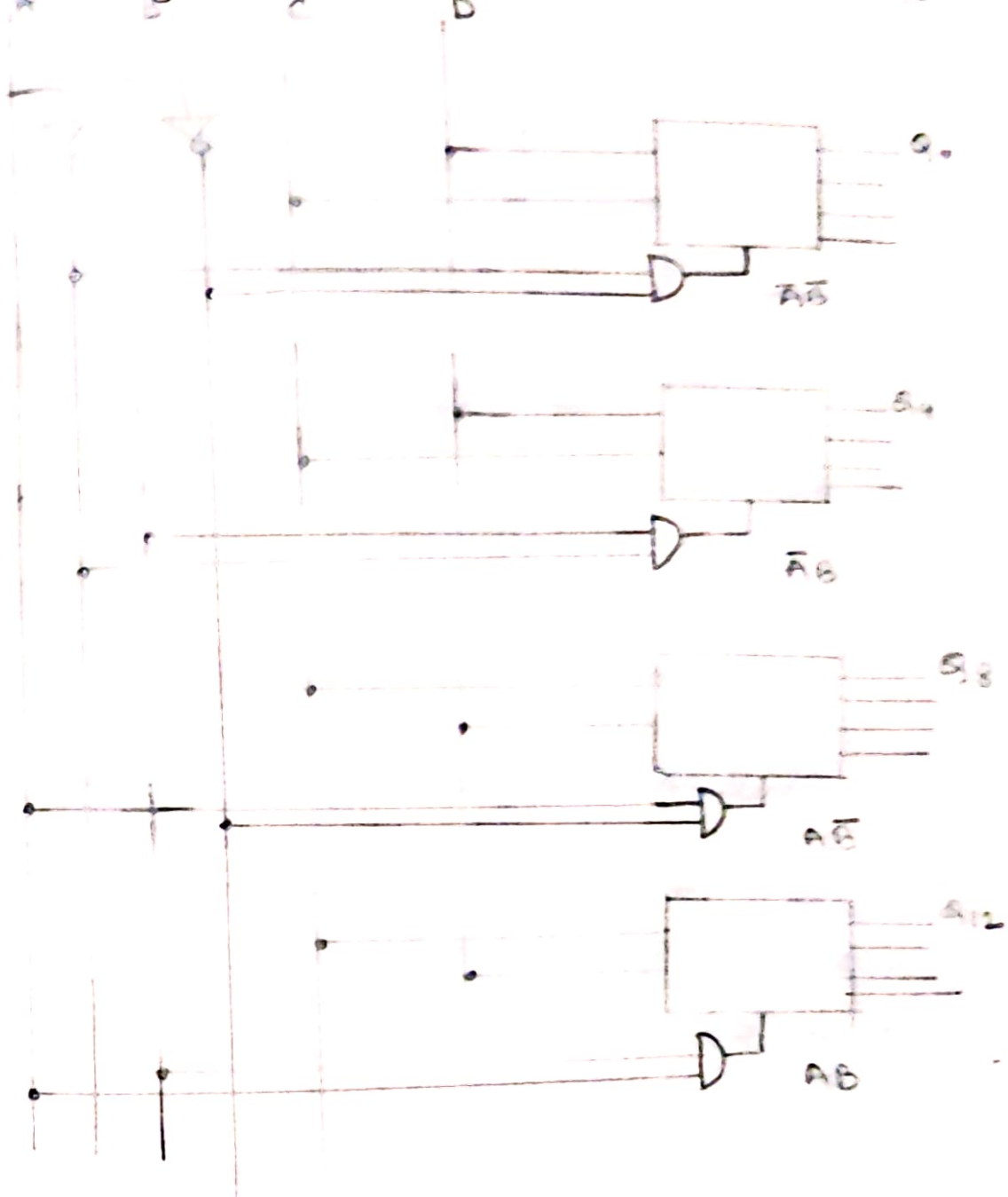


| Input |   |   | Output         |                |                |                |                |                |                |                |
|-------|---|---|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|
| A     | B | C | Q <sub>7</sub> | Q <sub>6</sub> | Q <sub>5</sub> | Q <sub>4</sub> | Q <sub>3</sub> | Q <sub>2</sub> | Q <sub>1</sub> | Q <sub>0</sub> |
| 0     | 0 | 0 | 0              | 0              | 0              | 0              | 0              | 0              | 0              | 1              |
| 0     | 0 | 1 | 0              | 0              | 0              | 0              | 0              | 0              | 1              | 0              |
| 0     | 1 | 0 | 0              | 0              | 0              | 0              | 0              | 1              | 0              | 0              |
| 0     | 1 | 1 | 0              | 0              | 0              | 0              | 1              | 0              | 0              | 0              |
| 1     | 0 | 0 | 0              | 0              | 0              | 1              | 0              | 0              | 0              | 0              |
| 1     | 0 | 1 | 0              | 0              | 1              | 0              | 0              | 0              | 0              | 0              |
| 1     | 1 | 0 | 0              | 1              | 0              | 0              | 0              | 0              | 0              | 0              |
| 1     | 1 | 1 | 1              | 0              | 0              | 0              | 0              | 0              | 0              | 0              |

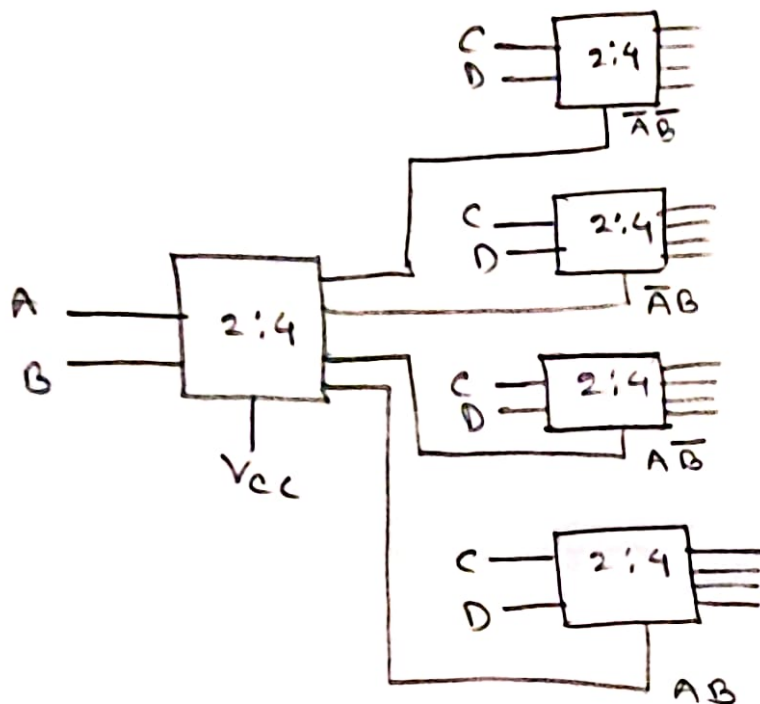
A=0 ~~000~~, 1st decoder enable 2221

A=1 " , 2nd " " " "

# Design 4:16 using 2:4 decoder.



# using only 2:4, design 4:16

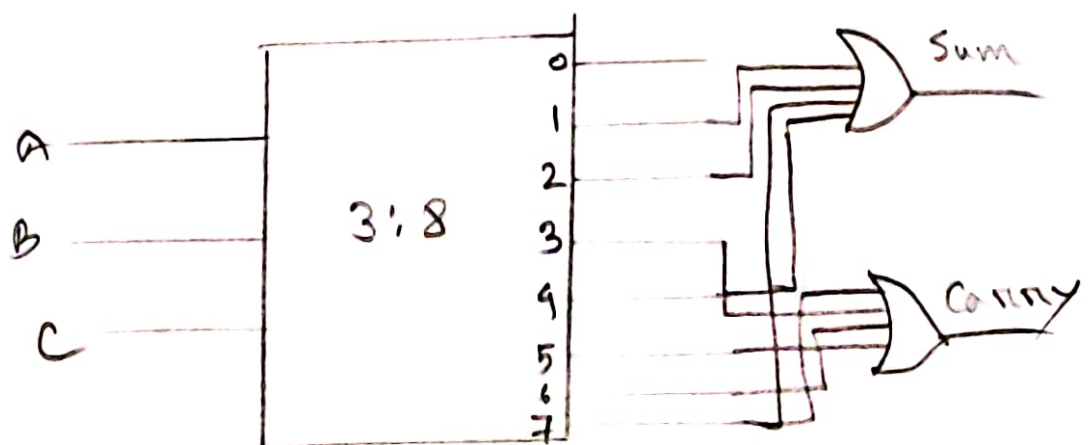


# Design full Adder using Decoder.

$$F(A, B, C) = \Sigma(1, 3, 5, 7)$$

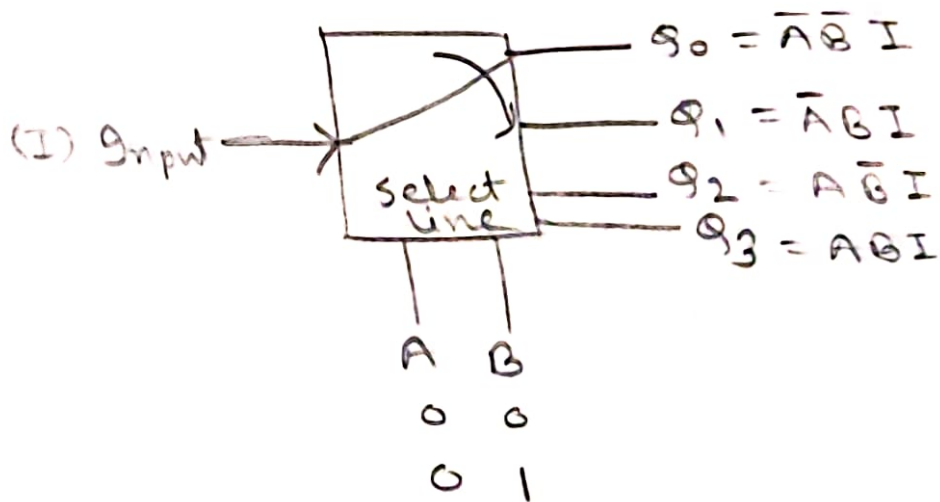
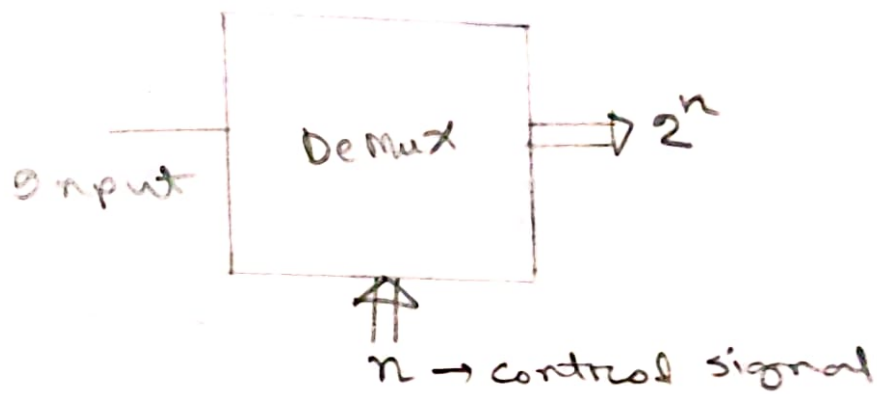
$$F_3(A, B, C) = \Sigma(1, 2, 4, 7)$$

$$F_C(A, B, C) = \Sigma(3, 5, 6, 7)$$

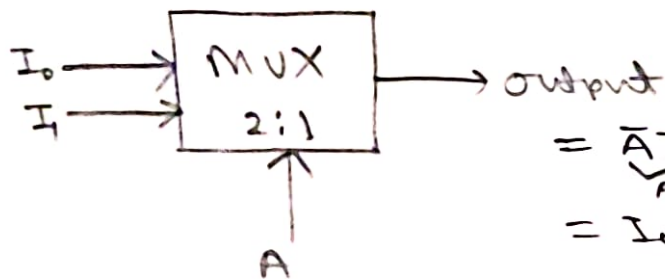
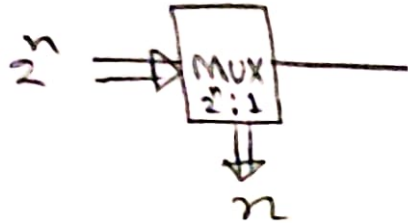


## Demultiplexer

एक single इनपुट को multiple आउटपुट में  
convert करता है।



## Multiplexer (MUX)



$$\begin{aligned} &= \underbrace{\bar{A}I_0}_{\text{AND}} + \underbrace{AI_1}_{\text{AND}} \\ &= I_0 \text{ OR } I_1 \quad \begin{matrix} | A=0 \\ | A=1 \end{matrix} \end{aligned}$$

$A=0$ ,  $I_0$  connected  
 $A=1$ ,  $I_1$  " "

★  $F(A, B, C) = \Sigma(0, 1, 2, 4, 6, 12)$   
using MUX.

06.12.2022

$$2^n = 4 \Rightarrow 2^{n+1} \text{ mux} \rightarrow \text{O/P}$$

function with a multiplexer  $\rightarrow$  book

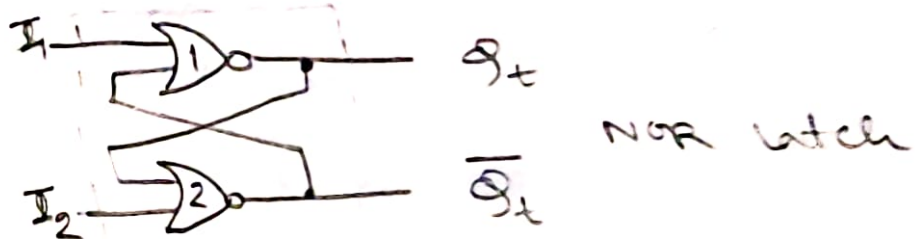
## Sequential Logic Circuit

present state depends on previous state.

### Flip-Flop

Latch  $\rightarrow$  memory element

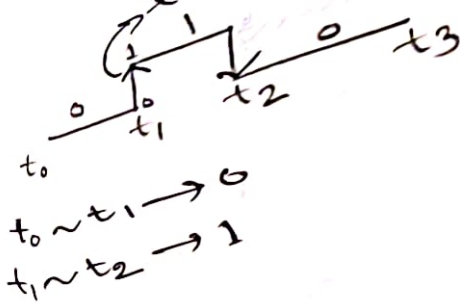
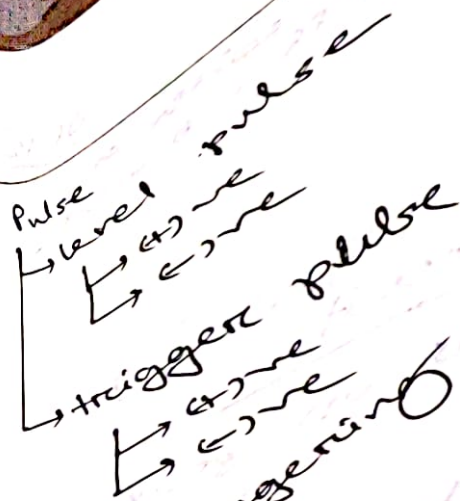
- $\rightarrow$  NOR Latch
- $\rightarrow$  NAND latch



- $\rightarrow$  RS FF (set-Reset)
- $\rightarrow$  D FF
- $\rightarrow$  JK FF
- $\rightarrow$  T FF

|           |       |       |                  |
|-----------|-------|-------|------------------|
|           | $I_2$ | $I_1$ |                  |
| for NAND, | 0     | 0     | } invalid / Race |
| for NOR,  | 1     | 1     |                  |

This problem arises in latch (SR)



→ level

ideal case  $\hookrightarrow$  0 থেকে 1 এ

কত time লাগছে না।

Real life এ perfect

square wave পাওয়া যায় না।

$t_0 \sim t_1 \rightarrow$  এটা কম use করা হয় as it takes some time

Rising time মত কমানা যায়, তত better.

Latch এ level pulse use করা হয়।

precision এ কয় non-linear use করতে হয়।

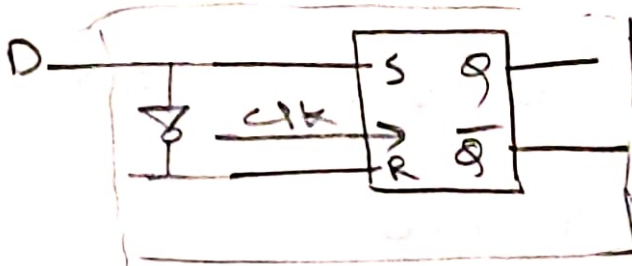
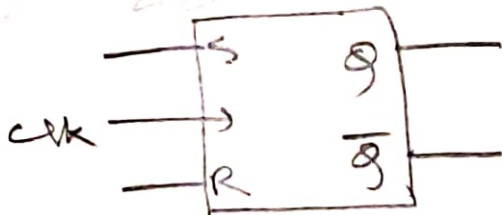
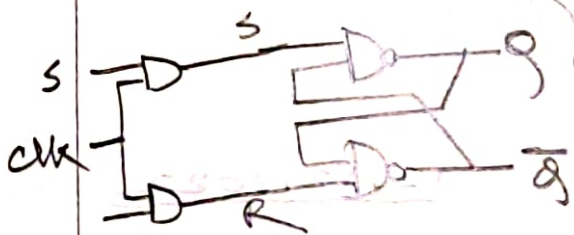
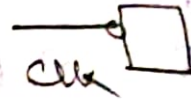
## Clock signals & clocked flip-flops

- Asynchronous
- Synchronous



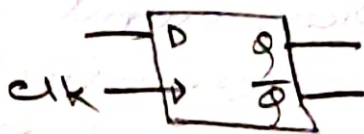
cascade  $\rightarrow$   $\overline{Q}$  input,  $Q$  output

$\rightarrow$  positive  
 $\rightarrow$  negative

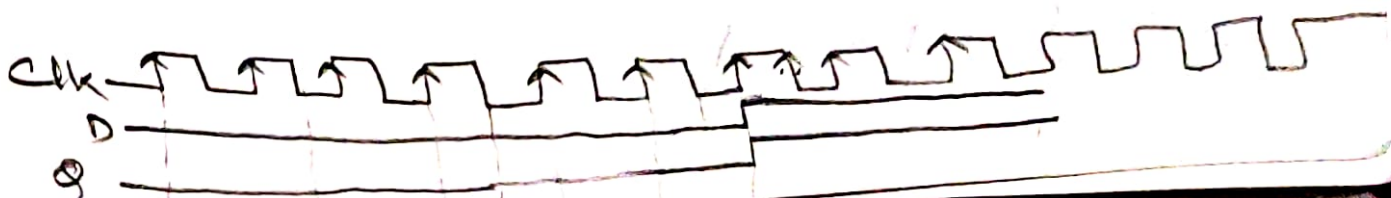


| clk        | S | R | Q       |
|------------|---|---|---------|
| No clock   | X | X | NC      |
| $\uparrow$ | 0 | 0 | NC      |
| $\uparrow$ | 0 | 1 | 0       |
| $\uparrow$ | 1 | 0 | 1       |
| $\uparrow$ | 1 | 1 | Invalid |

D - flip flop  $\rightarrow$  to transfer data



symmetry



D-flip flop is invalid output after it, but the problem is single input.

| clk | J | K | Q              |
|-----|---|---|----------------|
| ↑   | 0 | 0 | NC             |
| ↑   | 0 | 1 | 0              |
| ↑   | 1 | 0 | 1              |
| ↑   | 1 | 1 | $\overline{Q}$ |

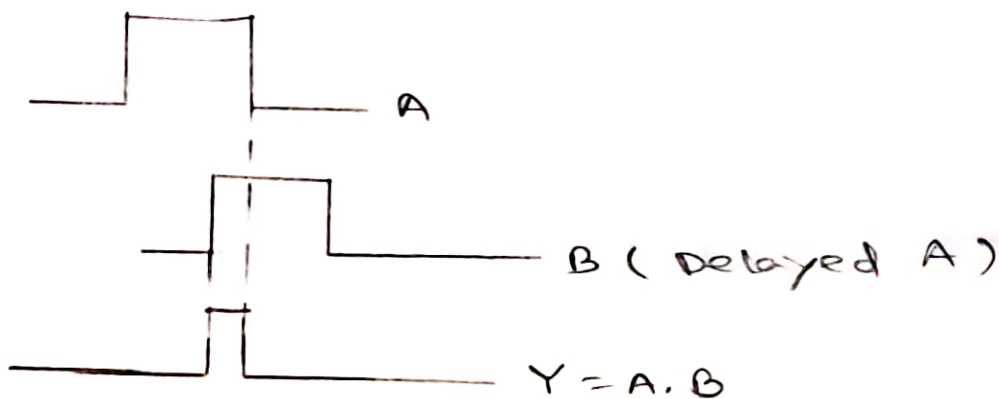
→ JK Flip flop

→ Toggle

12.12.2022

Clocked flip-flop is output waveform shape depend on clk as positive triggering or edge,

#

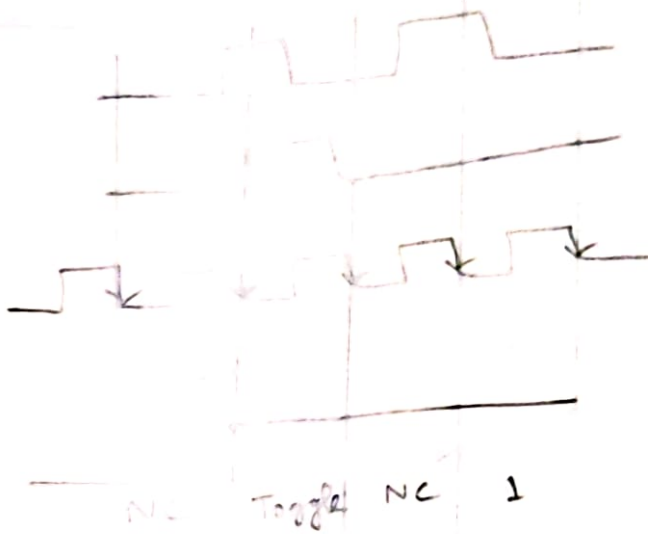


prev output

| prev output | $Q_t$ | D | $Q_{t+1}$ |
|-------------|-------|---|-----------|
|             | 0     | 0 | 0         |
|             | 0     | 1 | 1         |
|             | 1     | 0 | 0         |
|             | 1     | 1 | 1         |

prev. output is corner effect and clk is high!

#



JK  
→ clock

clocked D FF

| D | clk | Q |
|---|-----|---|
| 0 | ↑   | 0 |
| 1 | ↑   | 1 |

clocked T FF

| T | clk | Q      |
|---|-----|--------|
| 0 | ↑   | NC     |
| 1 | ↑   | Toggle |

★ Convert J-K into D mathematically.

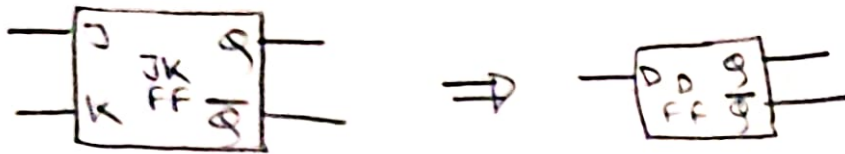
J-K

| J | K | Q         | $\bar{Q}$ |
|---|---|-----------|-----------|
| 0 | 0 | NC        | NC        |
| 0 | 1 | 0         | 1         |
| 1 | 0 | 1         | 0         |
| 1 | 1 | $\bar{Q}$ | Q         |

D

| D | Q | $\bar{Q}$ |
|---|---|-----------|
| 0 | 0 | 1         |
| 1 | 1 | 0         |

Excitation Table → to design input



Excitation Table of JK FF

| $Q_t$ | $Q_{t+1}$ | J        | K        |
|-------|-----------|----------|----------|
| 0     | 0         | 0,0<br>0 | 0,1<br>X |
| 0     | 1         | 1,1<br>1 | 0,1<br>X |
| 1     | 0         | 1,0<br>X | 1,1<br>1 |
| 1     | 1         | 0,1<br>X | 0,0<br>0 |

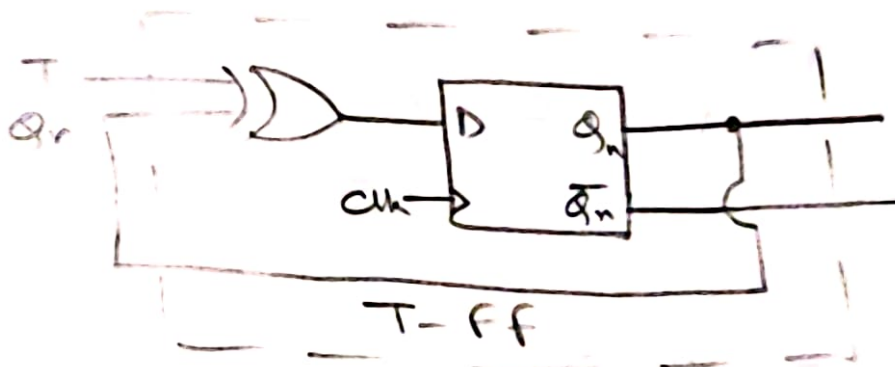
H.W Excitation of SR, D, T.

| $Q_t$ | $Q_{t+1}$ | SR       |          | D | T |
|-------|-----------|----------|----------|---|---|
|       |           | S        | R        |   |   |
| 0     | 0         | 0,0<br>0 | 0,1<br>X | 0 | 0 |
| 0     | 1         | 1        | 0        | 1 | 1 |
| 1     | 0         | 0        | 1        | 0 | 1 |
| 1     | 1         | 0,1<br>X | 0,0<br>0 | 1 | 0 |

### □ D-FF to T-FF

| I/P |       | O/P       |   |
|-----|-------|-----------|---|
| T   | $Q_n$ | $Q_{n+1}$ | D |
| 0   | 0     | 0         | 0 |
| 1   | 0     | 1         | 1 |
| 1   | 1     | 0         | 0 |
| 0   | 1     | 1         | 1 |

$$D = TQ'_n + T'Q_n = T \oplus Q_n$$



### □ J-K FF to D-FF

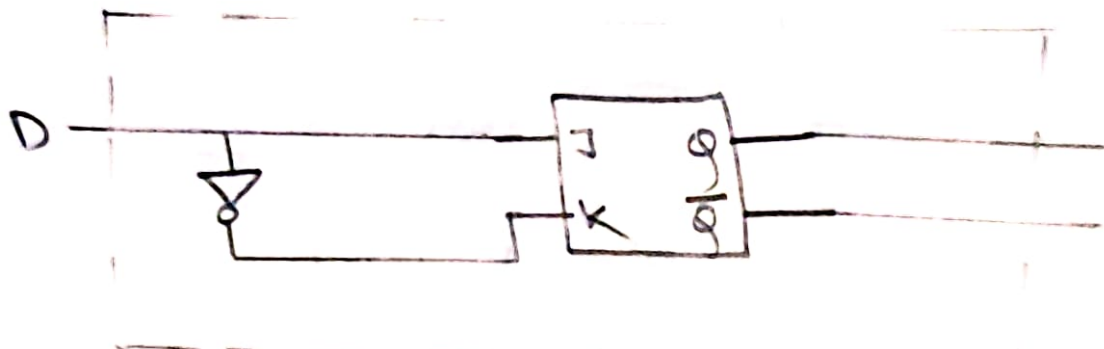
| I/P |       |           | O/P |   |
|-----|-------|-----------|-----|---|
| D   | $Q_n$ | $Q_{n+1}$ | J   | K |
| 0   | 0     | 0         | 0   | X |
| 1   | 0     | 1         | 1   | X |
| 0   | 1     | 0         | X   | 1 |
| 1   | 1     | 1         | X   | 0 |

| $Q_n$ |   |
|-------|---|
| 0     | 0 |
| 1     | 1 |

$$J = D$$

| $Q_n$ |   |
|-------|---|
| 0     | 1 |
| 1     | 0 |

$$K = \bar{D}$$



D-FF

RS-FF to D-FF

| D | $Q_n$ | $\bar{Q}_{n+1}$ | R | S |
|---|-------|-----------------|---|---|
| 0 | 0     | 0               | X | 0 |
| 1 | 0     | 1               | 0 | 1 |
| 0 | 1     | 0               | 1 | 0 |
| 1 | 1     | 1               | 0 | X |

$Q_n$

|     |   |
|-----|---|
| 0   | 1 |
| X 1 | 1 |
| 0   | 0 |

D

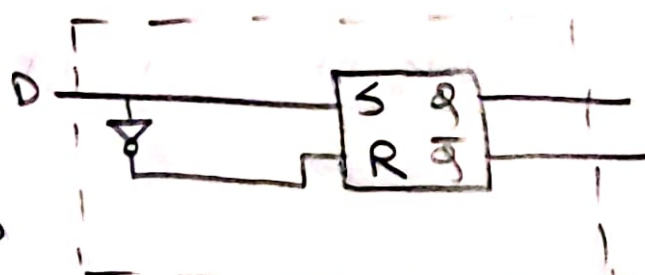
$$R = \bar{D}$$

$Q_n$

|   |     |
|---|-----|
| 0 | 1   |
| 0 | 0   |
| 1 | X 1 |

D

$$S = D$$



D-FF

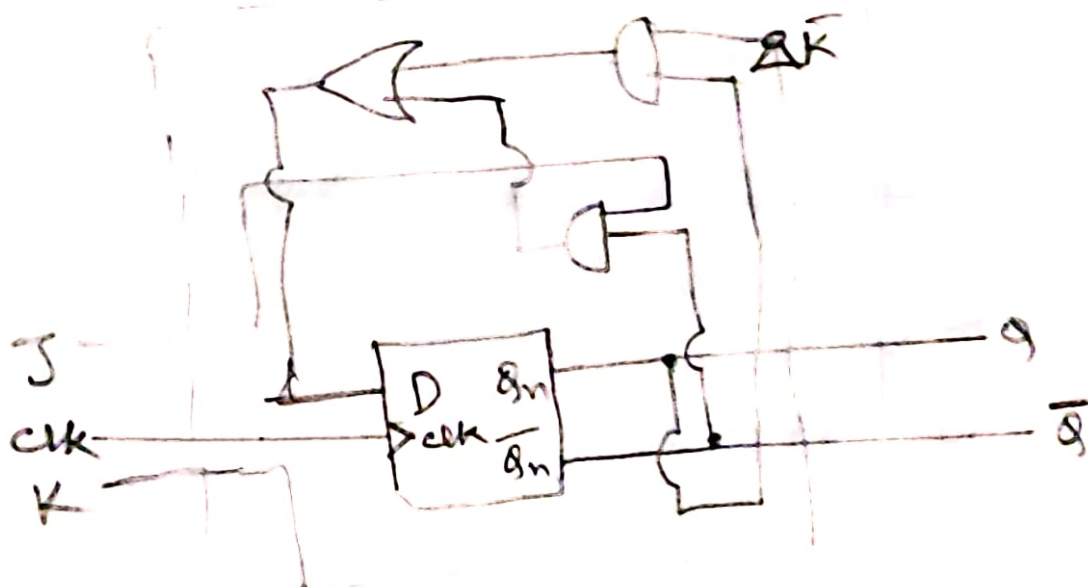
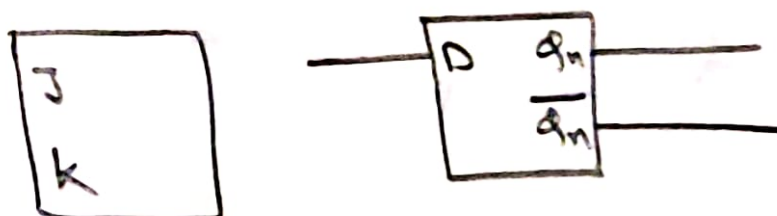
# Ex D-FF to JK-FF

| J | K | $Q_n$ | $Q_{n+1}$ | D |
|---|---|-------|-----------|---|
| 0 | X | 0     | 0         | 0 |
| 1 | X | 0     | 1         | 1 |
| X | 1 | 1     | 0         | 0 |
| X | 0 | 1     | 1         | 1 |

JK

|   | 00 | 01 | 11 | 10 |
|---|----|----|----|----|
| 0 | 0  | 0  | 1  | 1  |
| 1 | 1  | 0  | 0  | 1  |

$$D = J\bar{Q}_n + \bar{K}Q_n$$



## Counter

→ Asynchronous Counter / Ripple  
→ Synchronous "

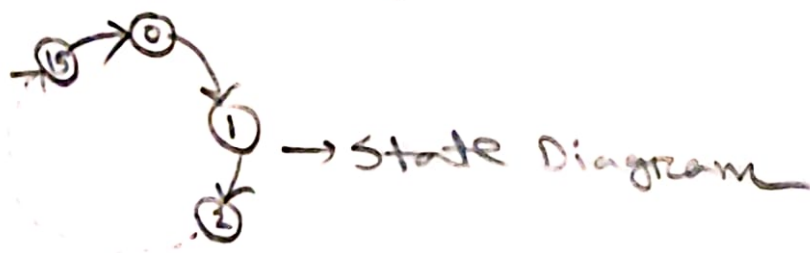
Counter হিসেবে highest (২) number count করে  
তার binary representation সমতুল্য bit  
সংখ্যে, সমতুল্য FF সংখ্যে।

JK FF এ  $J=1, K=1$  হলে, output frequency  $\frac{1}{2}$   
হয়ে যায়, so, 2 হিসেবে কাজ করে, তার binary  
counter.

Up counter  $\rightarrow 0, 1, 2, \dots, 15$

$$\text{output frequency} = \frac{\text{input frequency}}{2^n}$$

$n = \text{no. of flip-flop}$



MOD Number = no. of states before recycling



$T_{\text{clock}} < N \times t_{pd} \rightarrow$  Distorted output

proper  $\rightarrow T_{\text{clock}} > N \times t_{pd}$

# 5 w/o JK

latch  $\rightarrow 1 \text{ ns}$

NAND  $\rightarrow \frac{1}{2} \text{ ns}$

$\therefore \text{delay} = 5 \times (1 + \frac{1}{2}) = 7.5 \text{ ns}$

19.12.2022

Main problem of ripple/asynchronous counter  
 $\rightarrow$  time delay

$t_d =$  time delay for 1 FF

$$T = N \times t_d$$

$$f = \frac{1}{T} = \frac{1}{N \times t_d}$$

Synchronous/Parallel counters

All FFs  $\rightarrow$  simultaneously clk input to all

|                  |             |
|------------------|-------------|
| MSB $\leftarrow$ | LSB         |
| higher order     | lower order |

All lower order  $\rightarrow$  0 or 1, JK as input to all

1. Simultaneous clk pulse

2. 1st FF as input = 1

Other FF as input, lower order FF as output.

|   | A | B | C |                             |
|---|---|---|---|-----------------------------|
|   | 0 | 0 | 0 | $2^0 = 1$                   |
|   | 0 | 0 | 1 | $2^1$ position નો નો change |
| 4 | 0 | 1 | 0 |                             |
|   | 0 | 1 | 1 |                             |
|   | 1 | 0 | 0 |                             |
|   | 1 | 0 | 1 |                             |
|   | 1 | 1 | 0 |                             |
|   | 1 | 1 | 1 |                             |

LSB 1 Bit નો નો change કરતો રહે, એટલે  
toggle કરવાનો થાય JK કે 1 નો 2મો પ્રયોગ.

$$L = 2^N$$

$$\Rightarrow N = \lceil \log_2 L \rceil \rightarrow \text{No. of FF to design}$$

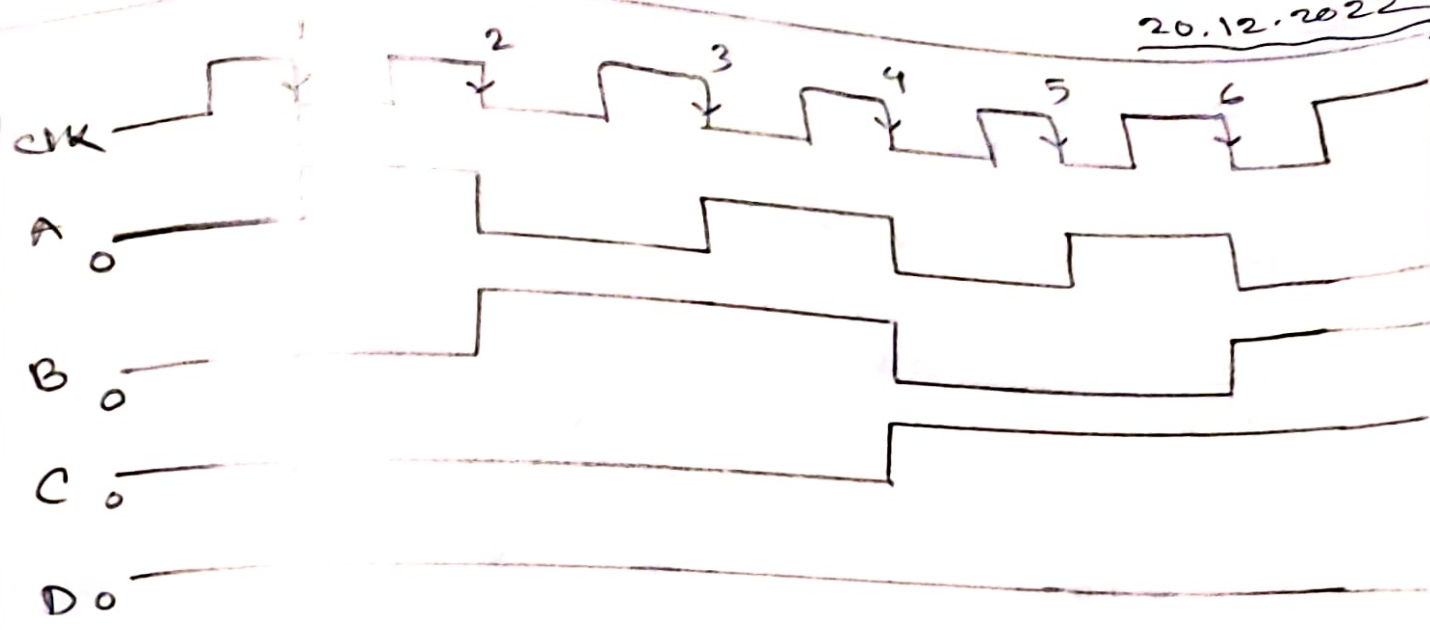
$$L = \text{No. of level.}$$

#Designing of synchronous counters:

- how many FF is needed.
- Rightmost કે LSB કે  $J = K = 1$
- સમુદાય same clk pulse
- lower order નો output નો and, નો  
પછી FF નો input.

Mod No.  $\rightarrow$  A counter counts how many distinct states.

20.12.2022



level = 16  
FF = 4

CLK → ~~data~~ change કરા  
input → ~~data~~ change કરા.

| MSB |   | LSB |   |
|-----|---|-----|---|
| D   | C | B   | A |
| 0   | 0 | 0   | 0 |
| 0   | 0 | 0   | 1 |
| 0   | 0 | 1   | 0 |
| ⋮   |   |     |   |

$2^N$  જાણી 0 થી back કરા 1  
 $k$  count કરા,  $2^N + (k-1)$  થી 0 થી 1  
 1 થી 0 થી 1 થી 0 થી 1

#  $0 \rightarrow 1 \rightarrow 3 \rightarrow 2 \rightarrow 1 \rightarrow 5$   
mod no. = 5

# 0 - 63 when 30?

→ 29 is the 30 count value

Counters with mod numbers  $< 2^N$

# 0 - 5

FF = 3

C B A

0 0 0

0 0 1

0 1 0

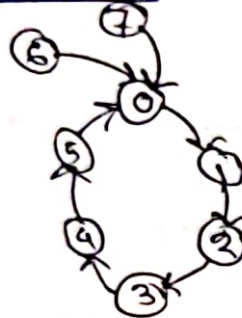
0 1 1

1 0 0

1 0 1

0 0 0

1 1 0 → 6



NAND = 0 → CLR input, forcefully 0  
Control Input

NAND as output CLR is used.

Up counter → increasing order

Down counter → Decreasing order

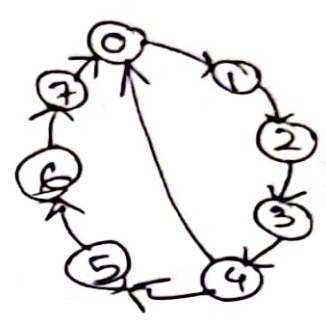
Down counter → output =  $\overline{Q}$

CT → lecture - 07

26.12.2022

# Analyzing Synchronous Counters

# 0 → 1 → 2 → 3 → 4



# (0-7) counter using D FF

BA

|    |   |   |
|----|---|---|
| 00 | 0 | 1 |
| 01 | 0 | 0 |
| 11 | 0 | 0 |
| 10 | 1 | 1 |

BA

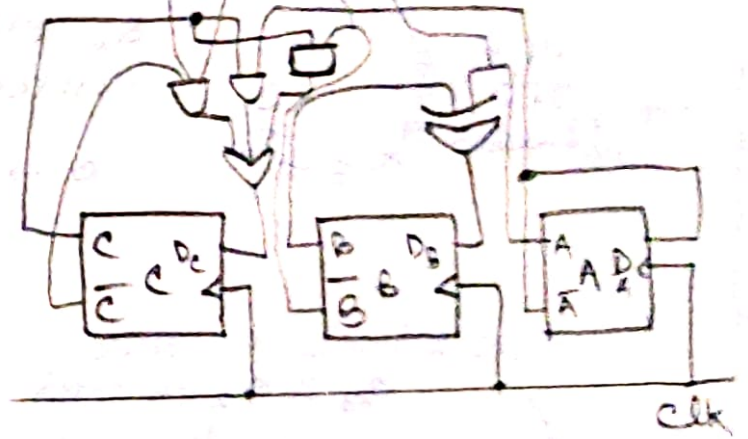
|    |   |   |
|----|---|---|
| 00 | 0 | 1 |
| 01 | 0 | 1 |
| 11 | 1 | 0 |
| 10 | 0 | 1 |

$$D_A = \bar{A}$$

$$D_B = D_C = \bar{B}C + AC + AB\bar{C}$$

C

|    |   |   |
|----|---|---|
| 00 | 0 | 0 |
| 01 | 1 | 1 |
| 11 | 0 | 0 |
| 10 | 1 | 1 |



$$D_B = A\bar{B} + \bar{A}B$$

$$= A \oplus B$$

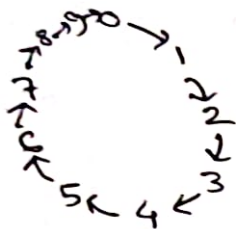
Transition diagram (226 present-next state fill 220 2201

27.12.2022

## IC Asynchronous Counter

IC  $\rightarrow$  74LS293

3-bit & 4-bit both are possible  
 $\hookrightarrow$  CP<sub>0</sub> a input n fir  
 #MOD 10



DCBA  
 1100  
 $\downarrow$

forcefully 0 કરો તો as couldn't be bigger than 9.

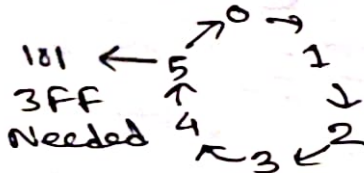
CLR = 0 તો, output = 0 [Reset]

Preset = 1 તો, output = 1 [set]

MR = Master ~~Reset~~ Reset

MR નો either MOD-8 or MOD-16 કરી શકો  
 (MR=0) (MR=1)

# MOD-6



Q<sub>0</sub> pin નાગર n CP<sub>0</sub> pin 6 નાગર n

0  $\rightarrow$  1  $\rightarrow$  2  $\rightarrow$  3  $\rightarrow$  4  $\rightarrow$  5  $\rightarrow$  6  $\rightarrow$  7  $\rightarrow$  8  $\rightarrow$  9  $\rightarrow$  10  
 110

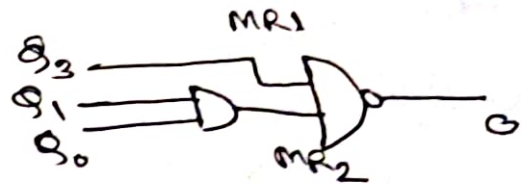
MR1 & MR2 તો કરી  
 Q<sub>3</sub> & Q<sub>2</sub> નો sort  
 તો જામણે n

Then MR1 = MR2 = 1 which will reset the FFs.

[તો તો bit નો forcefully 0 કરી શકો, તો bit જામણે તો MR નો connect કરી શકો]

## # MOD-11

|    | $Q_3$ | $Q_2$ | $Q_1$ | $Q_0$ |
|----|-------|-------|-------|-------|
| 0  | 0     | 0     | 0     | 0     |
| 1  | 0     | 0     | 1     | 0     |
| 2  | 0     | 1     | 0     | 0     |
| 3  | 0     | 1     | 1     | 0     |
| 4  | 1     | 0     | 0     | 0     |
| 5  | 1     | 0     | 1     | 0     |
| 6  | 1     | 1     | 0     | 0     |
| 7  | 1     | 1     | 1     | 0     |
| 8  | 0     | 0     | 0     | 1     |
| 9  | 0     | 0     | 1     | 1     |
| 10 | 0     | 1     | 0     | 1     |
| 11 | 0     | 1     | 1     | 1     |



একবার Reset হয়ে গেলে, আবার প্রথম থেকে count শুরু হয়।

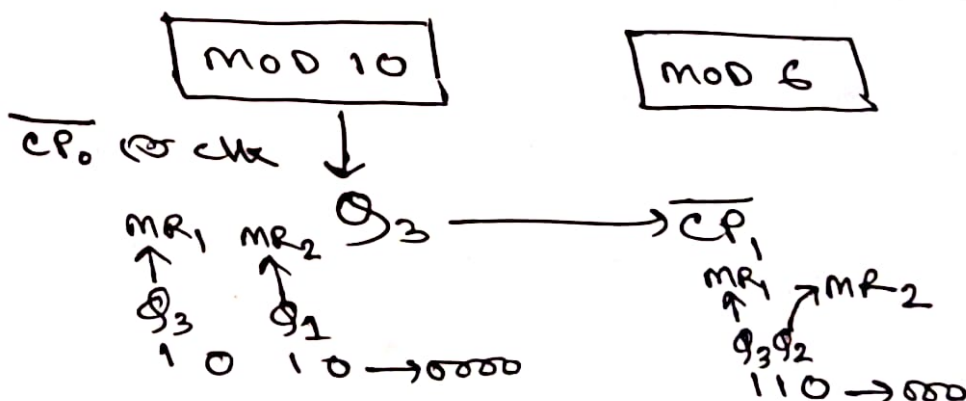
## \* MOD counter Timing Diagram

Limitation: mod-16 এর বেশি count করতে পারবে না।

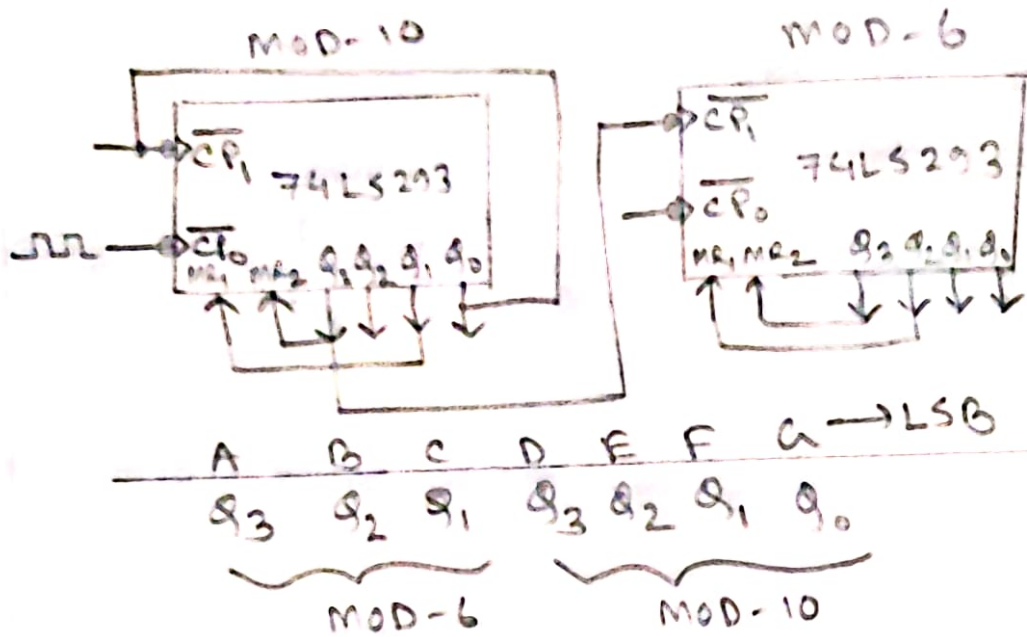
## # MOD-60 using 74LS293

→ **MOD-10**, **MOD-6** কে একত্রে series এ connect করতে হবে এর multiply করে 60 পর্যন্ত যাওয়া যায়।

6x10  
15x4 ওকাল  
মানে।



একটা IC এর last output, আনেকটা IC এর CP এ যাবে।



# MOD-58

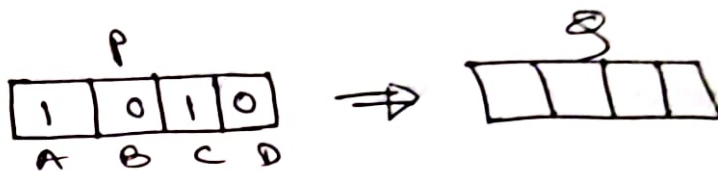
29 X 2

prime and and IC pin for and and

## # Register

Microprocessor ke first memory  
Temporary Storage Device

#



Total 48 registers hai, parallel  
registers kehte hai, series

## Classification

1. SISO (serial in, serial out)
2. SIPO (serial in, parallel out)
3. PISO
4. PIPO

Data Transfer ke liye best  $\rightarrow$  DFF

SISO ke liye clock pulse chahiye.

# 1010

After 12 pulse, what will be the output?

SIPO ke 28 pulse ke —

$4 \times 7 = 28$  pulse ke cycle chahiye.

After 4 pulse, 1010

$\therefore$  " 28 " , 1010

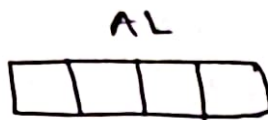
" 29 " , 0101 [1-bit ke transfer]

(4-bit)

PIPO ke cycle 1 pulse ke repeat 28 "

SIPO " " 4 "

# 4 bit registers



A = 1010

A to AL is store register, BL is transfer register

SISO → 8

SIP0 → 15 cases a register can pulse transfer?

PI SO

PI PO

12th pulse to AL & BL are value?

1st Pulse: AL 0 0 0 0

2nd " 0 1 0 0

3rd " 0 0 1 0

4th " 1 0 1 0

→ because of transfer, BL will take  
 after 12th pulse as BL is parallel connection

### Shift Register Counters

| Q <sub>3</sub> | Q <sub>2</sub> | Q <sub>1</sub> | Q <sub>0</sub> |
|----------------|----------------|----------------|----------------|
| 1              | 0              | 0              | 0              |
| 0              | 1              | 0              | 0              |
| 0              | 0              | 1              | 0              |
| 0              | 0              | 0              | 1              |
| 1              | 0              | 0              | 0              |



Right counter

↓  
 D is master Q<sub>0</sub> connect

Johnson Counter

↓  
 D is master Q<sub>0</sub>

02.01.2023

### Fan In

maximum number of inputs connected without voltage degradation.

### Fan out

maximum number loads that output of the gate can drive.

Fan in  $\uparrow$  & fan out  $\uparrow$ , expense  $\uparrow$

power dissipation  $\rightarrow$  only for DC (here)

Propagation delay  $\rightarrow$  average transition delay time

Noise immunity  $\rightarrow$  output voltage change at  $V_{OL}$ ,  $V_{OH}$  noise tolerate for  $V_{OL}$

noise margin  $\rightarrow$  quantitative measure of noise immunity

## Chapter-10

### TTL

Transistor transistor logic

BJT for gate design use

$\rightarrow$  TTL

power range  $10\text{mW}$  to  $10\text{W}$ ,  $V_{OL}$ ,  $V_{OH}$

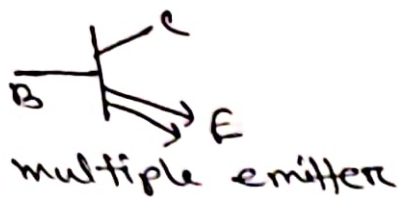
TTL use  $V_{OL}$ ,  $V_{OH}$



npn

MOS = Metal Oxide Semiconductor

CMOS  $\rightarrow$  unipolar, power dissipation low

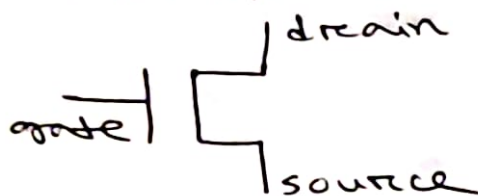


$$\# Y = a + bc$$

$$= P_1 + P_2$$

03.01.2022

MOSFET symbol



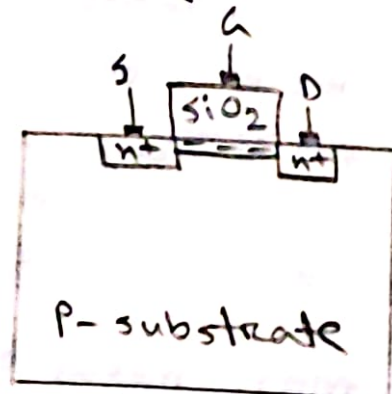
MOSFET as minority carrier as current leak  
 is not as ignorable. voltage is not as high.  
 power dissipation is less. precision better.

TTL  $\rightarrow$  BJT  $\rightarrow$  +12V / 15V / 18V for  
 exists as is.

MOSFET  $\rightarrow$  5V

$\hookrightarrow$  VLSI Technology

VLSI = Very Large Scale Integration



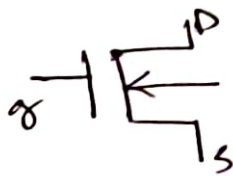
$n^+ \rightarrow$  huge concentration



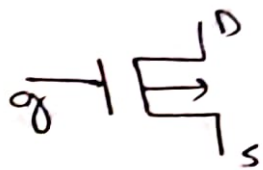
→ capacitors create noise

N-type substrate ( $10^6$  unit)

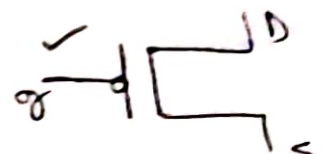
$N^+$  ( $>10^6$  unit)



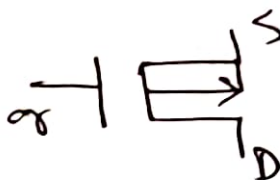
N-type MOS



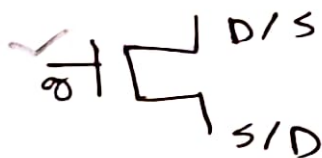
P-type MOS



P-type MOS



P-type



N-type

arrow always source to drain  
short circuit

✓ → Design for noise filter

gate voltage =  $V_g$

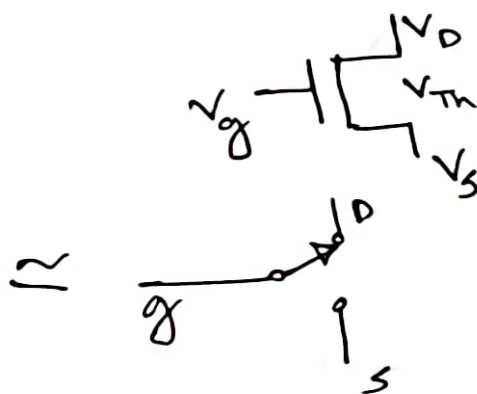
Drain " =  $V_D$

Source " =  $V_S$

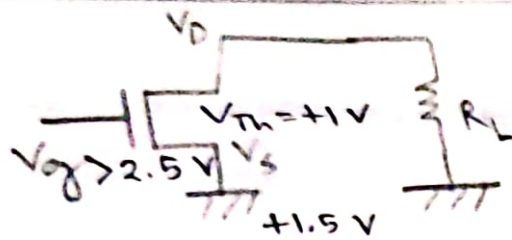
MOSFET/TTL → switch  
दिले का करे करे

$V_{Th}$  = Threshold voltage

$V_{Th} = +1V$  [Typically]



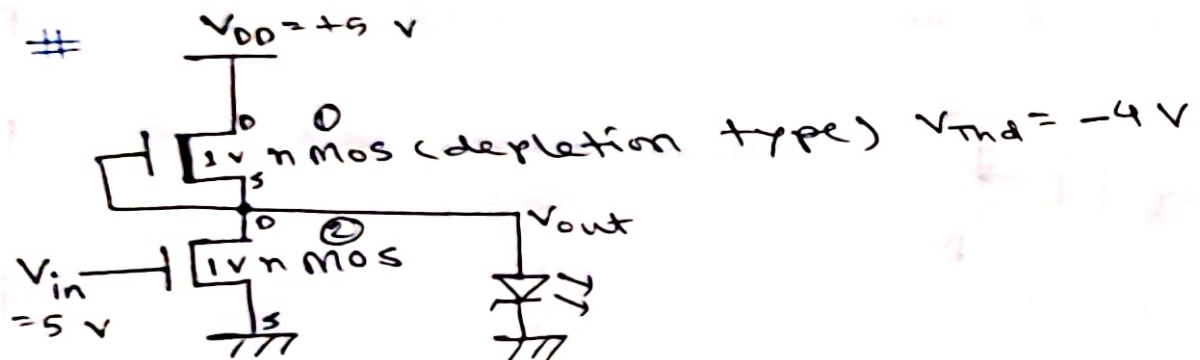
control input → g



$$\text{Input voltage} = V_{gs} = V_g - V_s$$

Turn on condition  $\rightarrow V_{gs} \geq V_{th}$

Turn off  $\rightarrow V_{gs} < V_{th}$



$V_g = V_s \rightarrow$  depletion transistor a short circuit  
 $\Rightarrow V_{gs} - V_s = 0$

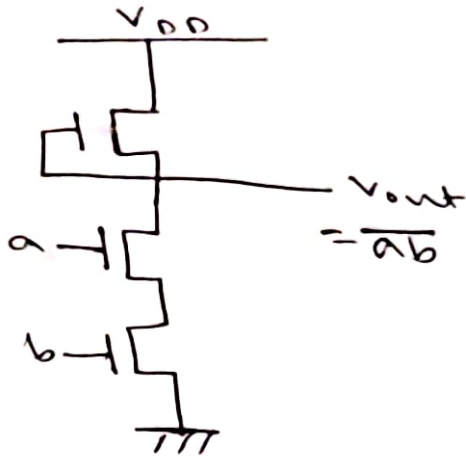
maximum voltage  $\rightarrow$  drain

② No. transistor on  $\rightarrow$  low resistance path  
 (current conduction 1  $\rightarrow$  2  $\rightarrow$  LED  $\rightarrow$  ON)

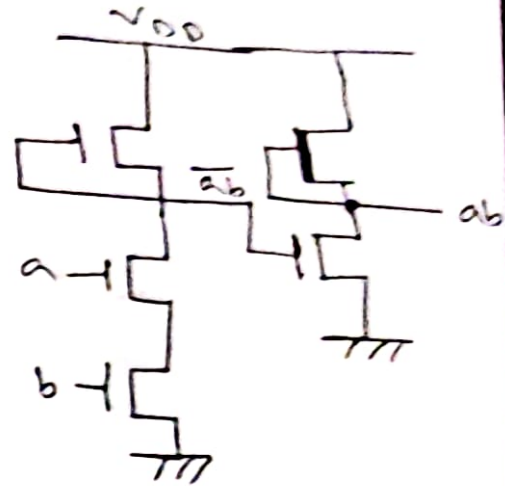
also  $V_{in} = 0$   $\rightarrow$  LED OFF

So,  $V_{in} = 5V$ , LED OFF  
 $V_{in} = 0V$ , LED ON } works like an inverter.

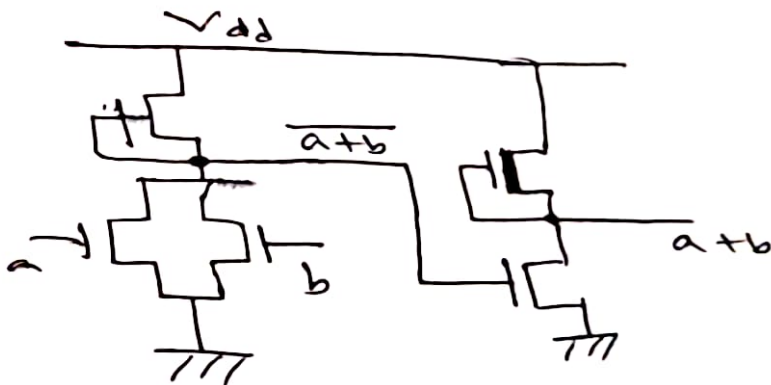
#  $Y = \overline{ab}$



#  $Y = ab$



#  $Y = a + b$



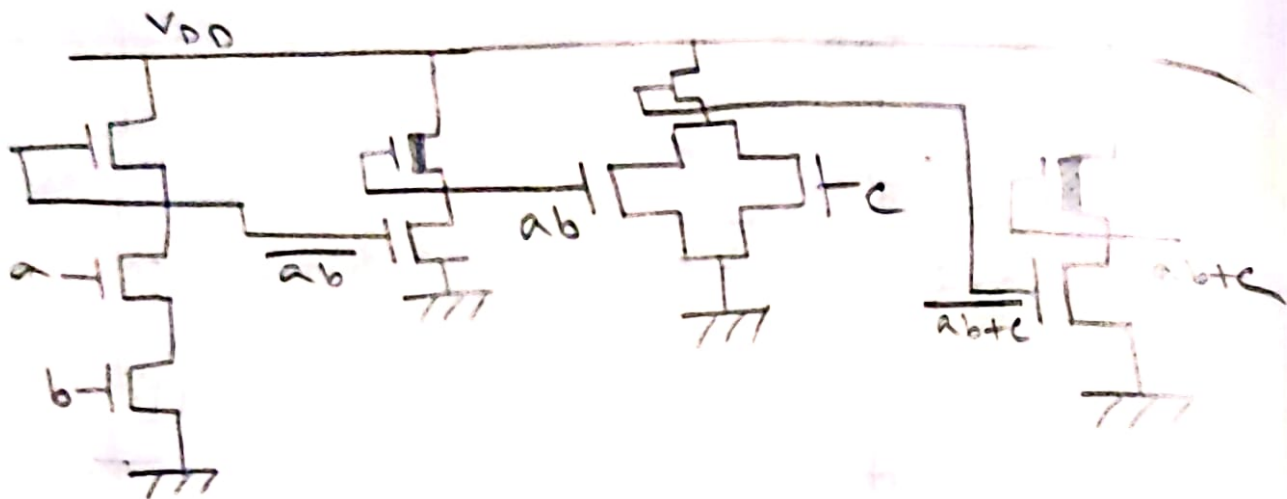
MOSFET

- Depletion Type (Predefined Threshold)
- Enhancement Type

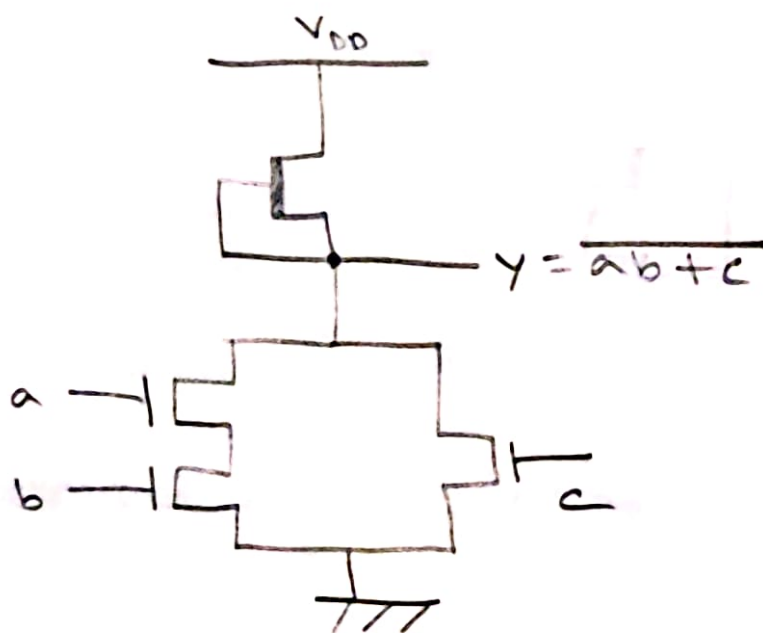
series → NAND

parallel → NOR

#  $Y = \overline{ab+c} = \overline{p_1+c}$ , where  $p_1 = ab$

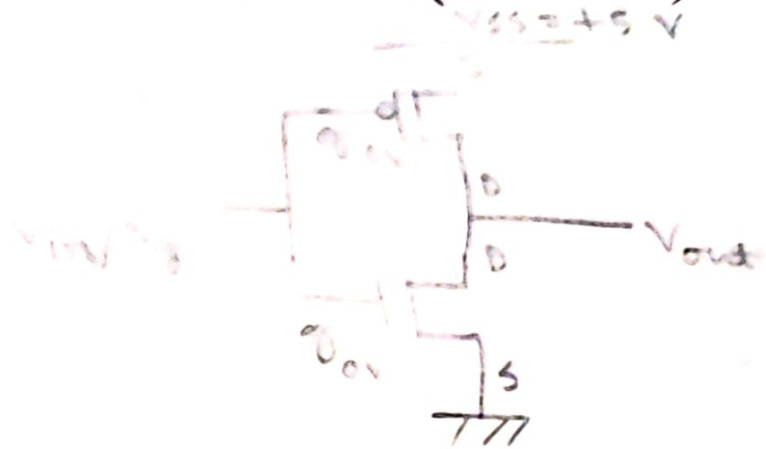


Simplified,



transistor on (cra, current is path from  
 1000 1

CMOSFET  $\rightarrow$  Complementary MOSFET



$$V_{in} = 0V \text{ (LOW)}, \\ V_{sg} = 5V$$

$$V_{gs} = 0V$$

$$\therefore V_{out} = 5V$$

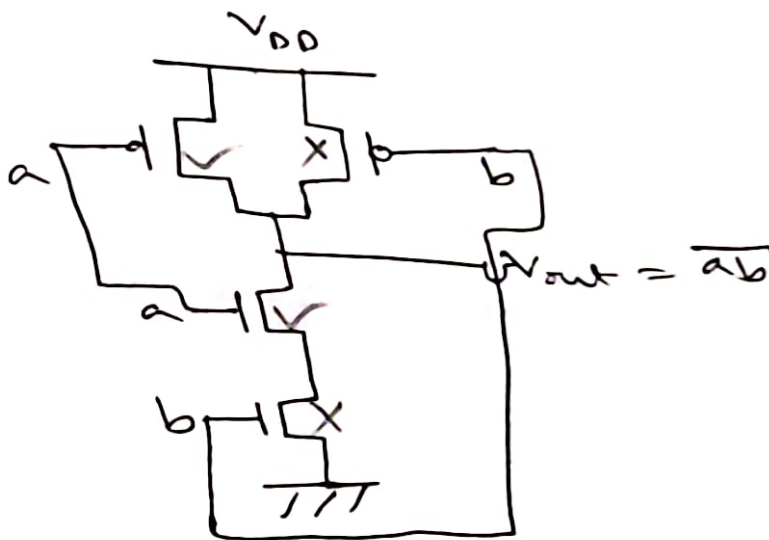
$$V_{in} = 5V \text{ (HIGH)}, \\ V_{sg} = 0V$$

$$\therefore V_{out} = 0V$$

$\therefore$  CMOS is itself an inverter.

CMOS = PMOS + NMOS

#  $y = \overline{ab}$  using CMOS.



NAND  $\rightarrow$  P: Parallel, N: series

NOR  $\rightarrow$  P: series, N: Parallel

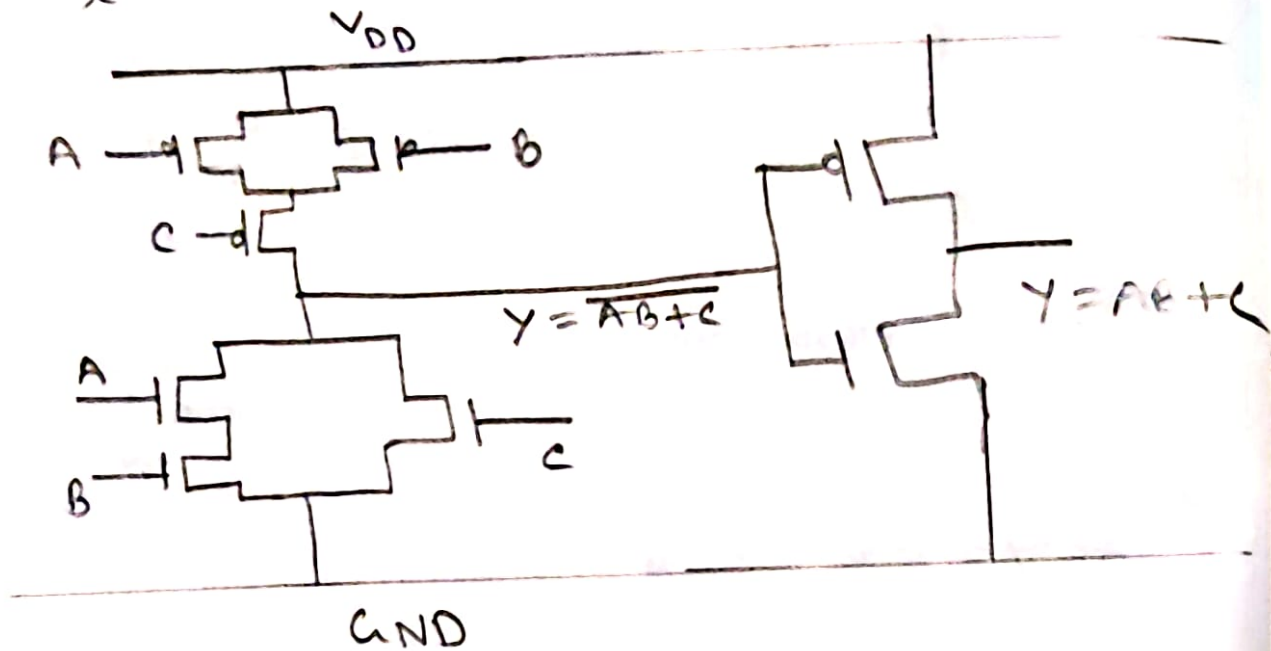
09.01.2022

# 4 bit asynchronous 120 kHz

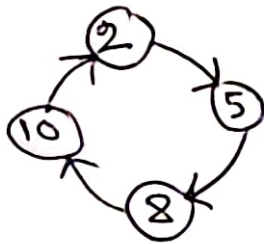
$\therefore Q_2 \rightarrow 15 \text{ Hz}$

NMOS  $\rightarrow$  Switch

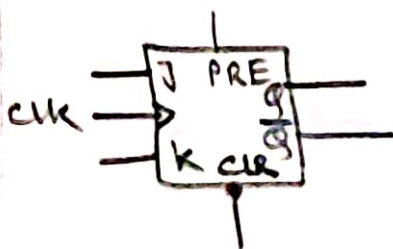
#  $Y = AB + C$  using CMOS.



#



Soln  $\rightarrow$  present state  
2p aspect  $\rightarrow$



PRESET, Clear  
 $\rightarrow$  control input

| Present State |   |   |   | Next State |   |   |   |
|---------------|---|---|---|------------|---|---|---|
| A             | B | C | D | A          | B | C | D |
| 0             | 0 | 1 | 0 | 0          | 1 | 0 | 1 |
| 0             | 1 | 0 | 1 | 1          | 0 | 0 | 0 |
| 1             | 0 | 0 | 0 | 1          | 0 | 1 | 0 |
| 1             | 0 | 1 | 0 | 0          | 0 | 1 | 0 |

Either preset as 1 or clear as 0.

Preset as 2 is just connect to +5 V for  
 2 active state, 0 & 1 are ignored.

$$J_A = ? \quad K_A = ?$$

Input  $\rightarrow$  present state

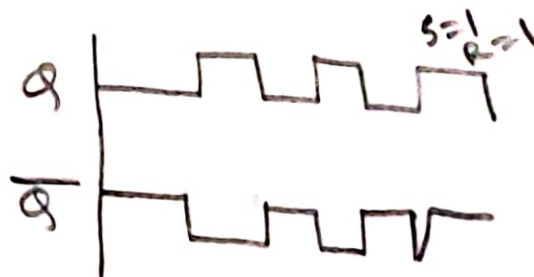
output  $\rightarrow J, K$

is a sol<sup>n</sup> for the problem.

Table is a 3-state (15) flip, k-map for  
 the state.

Random Counter synchronous flip design  
 for 1.

$$\# S=1 \\ R=1$$



| A | B |                |
|---|---|----------------|
| J | K | Q <sub>1</sub> |
| 0 | X | 1              |
| 1 | X | 0              |
| X | 0 | 0              |
| X | 1 | 0              |